BANK DATA PRIVACY PRESERVATION SYSTEM USING HOMOMORPHIC ENCRYPTION

Major Project

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INTRODUCTION

- Digital banking offers convenience but raises serious data privacy and security risks due to cloud storage.
- Sensitive financial data stored on third-party servers is vulnerable to breaches, insider threats, and unauthorized access.
- Traditional encryption protects data at rest and in transit, but requires decryption during computations, exposing data temporarily.
- Homomorphic Encryption (HE) enables computations directly on encrypted data without revealing actual values.
- The project SecureBank demonstrates this using the Paillier scheme, integrated with Firebase and a Flask-based API for secure cloud banking.

LITERATURE REVIEW

	Published Date	Technology Used	Datasets Used	Performance Metrics	Remark
1	IEEE Access, Vol. 9 · Published July 2021	Fully Homomorphic Encryption (FHE) Homomorphic Authenticators (FHA + OR-homomorphic) Lattice Trapdoors (SIS hardness) Encrypt-then-Authenticate paradigm Pseudorandom Functions (PRF)	Theoretical / no empirical dataset	IND-CCA security · SUF-CCA security · Adaptive security · Amortized efficiency · Ciphertext compactness	First construction of fully homomorphic authenticated encryption (FHAE). New unified "all-in-one" security notion stronger than prior separate privacy + authenticity definitions. Supports multiple datasets. Based on real-or-random formalism instead of find-then-guess.
2	IEEE Trans. Signal & Info. Process. Over Networks, Vol. 10, 2024 · Published May 2024	State Decomposition Gradient Tracking (Push-Pull) Paillier Cryptosystem (PcPPSD extension) DLG Attack Testing	Cleveland Heart Disease (UCI, 297 samples, 14 features) MNIST (60,000 samples, 784 features)	Optimality gap · Convergence rate · Privacy metric Di(IA(0:k)) · MSE (DLG) · Convergence time	First use of state-decomposition in decentralized optimization for gradient privacy. No accuracy–privacy trade-off unlike differential privacy. Honest-but-curious attacker model. PL condition (non-convex) supported.
3	IEEE Journal of Biomedical & Health Informatics, 2023	Secret sharing (Lagrange) Weight mask (PRG + DH) Digital signature + hash CNN training Periodic update strategy	MNIST CIFAR-10 Vehicle dataset	• Accuracy: 96% (MNIST), 65% (CIFAR-10) • Training time: ~5000s vs 100k+ s (FedAvg) • Energy: 1000–100,000 J (vs 10k–700k J) • 40% faster than differential privacy • Anti-gradient-leak: image not recoverable	PPFLEC is a lightweight edge-FL scheme for IoMT. Avoids homomorphic encryption overhead via secret sharing + mask. Periodic update reduces communication rounds by 1/τ. Demonstrated resistance to DLG gradient leak attacks. Comparable accuracy to FedAvg without privacy cost.
4	IEEE Trans. Dependable & Secure Computing, 2020	Paillier PHE (HRES) KP-ABE Diffie-Hellman key exchange Two-level decryption	Simulation (synthetic)Java jPBC library	• Computation time (ms) per operation • 7 operations: Add, Sub, Mul, Sign, Abs, Compare, EqualTest • Communication overhead (bits) • Complexity: O(1) most ops, O(N) for Multiplication	First scheme combining PHE-based computation (7 ops) with ABE-based fine-grained multi-user access control. Outperforms prior single-user PHE/FHE systems. Suitable for big data and IoT analytics. DR decryption cost ~25ms at 2048-bit n.
5	IEEE Trans. Information Forensics & Security, 2022	CKKS (FHE) Cosine similarity Blockchain / Smart contract ReLU under ciphertext 2-party computation	MNIST FashionMNIST	• Test accuracy (vs FedSGD, Krum) • Test error rate under 10%–50% malicious clients • Encryption/decryption time (vs Paillier, BatchCrypt) • Gas cost per iteration • Convergence bound: $\|w_i - w^*\|$	PBFL resists both targeted and untargeted poisoning attacks. Uses root dataset + cosine similarity for gradient filtering. Blockchain ensures transparency. CKKS outperforms Paillier in efficiency for large-scale vectors.

	Published Date	Technology Used	Datasets Used	Performance Metrics	Remark
6	IEEE Access, 2020	LWE problem GSW scheme CLFHE SamplePre / SampleLeft Random oracle model	Simulation onlyNo real-world dataset	• Semantic security (INDr-ID-CPA) • Anonymity under Type I & II attacks • Circuit depth L support • Ciphertext compactness	First CLFHE supporting both homomorphic addition and multiplication. Proves security in both random oracle and standard models. No key escrow problem. Future work includes ring-LWE and CCA security.
7	Tsinghua Science & Technology, 2022	MKHE / CDKS19 CKKS17 RLWE Rescaling Gadget-free relinearization	Neural network image classification (CDKS19 benchmark)	• Eval key size: R^{13} (vs R^d^3) • Error variance: $k^2n^2\sigma^2$ (vs $k^2n^2d^2\sigma^2V^2g$) • Computation cost: $2k^2n^2$ multiplications • IND-CPA secure	Optimizes CDKS19 MKHE by eliminating gadget vectors via modulus elevation + rescaling. Significantly reduces evaluation key size and error growth. Suitable for multiuser cloud computing and AI applications.
8	IEEE Networking Letters, Vol. 5, No. 4 · December 2023	Paillier Partial HE (PHE) Federated Learning (FedAvg) MLP Neural Network SGD Optimizer 5G NWDAF Architecture	MNIST (70,000 handwritten digit images, 28×28 px)	Classification accuracy · Execution time per round per client · Key length vs. time (128/256/512/1024 bit)	Novel application of PHE to secure 5G/6G NWDAF federated learning. Deters gradient leakage attacks. HE computation overhead is significant but feasible in core-network deployments. Advantage over DP: no accuracy degradation. Separation of key server and FL server enhances cross-domain privacy.
9	Tsinghua Science and Technology, Vol. 19, No. 5 · October 2014	Discrete Logarithm Problem (DLP) Double Decryption Algorithm (Dec1 + Dec2) ElGamal-inspired group cryptography Pollard's Lambda Method 3-component ciphertext (c1, c2, c3	Theoretical / no empirical dataset	Semantic security (IND distinguishing game) · Targeted / full homomorphic property · Decryption time $O(\sqrt{n})$ for Dec2	First and most efficient targeted FHE scheme at publication. No bootstrapping or relinearization required unlike Gentry/Brakerski schemes. Limitation: only semantic security proved; CCA1 security remains open. Dec2 requires discrete log computation – suitable for short messages via digital envelope only.
10	May 22, 2025 (pub) Jul 11, 2025 (current ver.)	CKKS FHE Scheme FPGA (Xilinx ZCU102, U280, Arria 10) HLS (Vitis 2023.1) NTT / INTT Barrett Modular Reduction (optimized) Design Space Exploration (DSE) Microsoft SEAL (RNS / SIMD)	HEXL Benchmarks (matrix mult) LoLa-MNIST LoLa-CIFAR	NTT: 1.50× speedup vs HEAX Rotation: 1.16× speedup LoLa-CIFAR: 6.09× vs FxHENN (ZCU102) LoLa-MNIST: 1.26× vs FxHENN Energy: 1.15–5.53× improvement DSP reduction: 18→4 units at 54-bit	Proposes full FPGA design automation framework for CKKS. Jointly optimizes HE parameter selection and hardware resource allocation. Key innovations: optimized Barrett reduction (DSP-LUT balance), parallel NTT-form automorphism, on-/off-chip memory co-management with HP port scheduling.

	Published Date	Technology Used	Datasets Used	Performance Metrics	Remark
11	Jun 8, 2021 (pub) Feb 18, 2022 (current ver.)	Blockchain (RAFT consensus) Federated Learning Paillier HE (additive) Differential Privacy (Laplace) K-means, Random Forest, AdaBoost Apache Spark distributed cluster	UCI Adult Dataset (income census data)	F1-Score: up to ~0.80 Accuracy: up to ~76.25% Faster runtime vs. single-node Improved clustering vs. traditional K-means	Combines blockchain + federated learning + dual privacy (DP + HE) for IIoT data security. Addresses model extraction and reverse attacks. Three ML algorithms each protected by different privacy methods. Improved Paillier (g = n+1) for efficiency.
12	11th ICCCNT 2020, IIT Kharagpur · July 2020	Paillier Homomorphic Encryption Proactive Secret Sharing Digital Signature (verifiability) Additive & Subtraction Homomorphism	Custom numerical secrets (implementation demo) No external benchmark dataset	Share computation time vs. bit size · Execution time vs. number of shares	Proposed proactive secret sharing with freedom to choose own share. Non-deterministic Paillier ensures same-content shares produce different ciphertexts. Supports threshold, proactive, and verifiable properties simultaneously – a first among surveyed schemes.
13	Jan 3, 2025 (pub) Jul 2025 (current ver.)	RLWE-based HE (CKKS) SIMD / Batching Secure SIFT BSDP, BSSP, BSCP, BSRP, BSIP protocols PALISADE Library (C++)	INRIA Graffiti Dataset Caltech-256 Dataset Oxford 5K Database BossBase (600 images)	PSNR & SSIM (visual security) Recall vs. 1-Precision Keypoint ratio ≈1.0 8/10 correct matches Computation time: ~134s (200×200 img)	Proposes word-wise protocols for fixed-point numbers avoiding binary conversion; achieves higher accuracy and efficiency than state-of-the-art privacy-preserving SIFT schemes (Hu et al., Jiang et al.). 128-bit security level.
14	April 2021 IJECE Vol. 11	Paillier PHE (additive) RSA (multiplicative) Proxy re-encryption Cloud storage + proxy server	Synthetic banking DB 5 customer records (name, address, account). 1 KB & 10 KB test sizes. Self-created experimental data.	128-bit key: Enc=0.004s, Re-enc=0.002s 2048-bit key: Enc=0.241s, Re-enc=0.006s Intel E5-2680 V3, 2.5 GHz, 1 GB RAM (server)	Practical proposal for securing banking data in the cloud using Paillier additive HE combined with a proxy re-encryption server for double-layer security. Shared public key generated from both storage and proxy servers. Computations (add/subtract) performed on ciphertext without decryption. Suitable for balance inquiries and transactions.
15	2015 IJSIA Vol. 9, No. 6	RSA (multiplicative HE) Paillier (additive HE) DGHV (somewhat HE) Hybrid HE scheme Citrix cloud platform + VPN	Synthetic banking accounts Customer names, account numbers, deposit amounts (<1 KB). Also tested 10 KB & 100 KB text for RSA/Paillier/DGHV benchmarks.	RSA 1024-bit: Enc=2.38s, C1×C2=11ms, Dec=2.37s (1KB) Paillier 512-bit: Enc=8.67s, Add=71ms, Dec=17s (1KB) DGHV 64-bit: Enc=44s, Add=8.5s (128B only) RSA fails at 100KB + 1024-bit	Hybrid approach: RSA for multiplicative keyword search on account names/IDs; Paillier for additive computation of deposit totals. Compared RSA, Paillier, and DGHV across key sizes and data sizes. DGHV too slow for practical banking. RSA + Paillier combination at 1 KB scale is practical. Recommends managing private keys with strong backup systems – key loss = permanent data loss.

	Published Date	Technology Used	Datasets Used	Performance Metrics	Remark
16	Jul 30, 2025 (pub) Nov 2025 (current ver.)	Regulator-Based Encryption (RBE) Q-Problem (post-quantum) Multiplicative neutral element Ring Z_P arithmetic Grover's algorithm resistance	1000 synthetic random messages (128–512 bit lengths) Raspberry Pi (edge device)	Entropy: 7.27 bits/symbol (vs RSA 7.13, NTRU 3.32) Complexity: $O(nl^2)$ vs $O(n^3)$ Encryption: 4.375ms (laptop), 7.93ms (RPi) HMult: <1.2ms on RPi	Novel FHE scheme tailored for IoT/edge devices with post-quantum security. Outperforms RSA, AES, NTRU in entropy and ciphertext uniformity. Parametrizable vector length (l=4–32 recommended). First to combine FHE + Q-Problem hardness.
17	Apr 2, 2024 (pub) Apr 11, 2024 (current ver.)	LWE-based FHE (GSW variant) Identity-Based Encryption (IBE) Multi-Key FHE (MKFHE) Decomposition Method Trapdoor functions Gadget matrix	Theoretical / formal No empirical dataset	Noise expansion: reduced to constant K (from poly(n,ℓ)) Aux. ciphertext: $\tilde{O}(n^2L^4)$ vs $\tilde{O}(n^4L^4)$ No fresh ciphertext needed IND-sID-CPA secure (LWE) Dimension: 2n log q	Introduces "decomposition method" for MKFHE eliminating pre-generation of fresh ciphertexts. Significantly reduces auxiliary ciphertext scale and noise growth. Smaller modulus q enables efficiency gains. Security proven in random oracle model. Outperforms schemes in [6],[21],[22].
18	July 2023 IEEE Micro	ElGamal HE (EC) Edwards25519 ECC FPGA (Intel Agilex AGF027) SecureBoost / XGBoost Federated Learning (Vertical)	Epsilon (250K×2000) Higgs (200K×30) Bank (40K×48) Credit (30K×23) 4 public benchmark datasets	6.1× speedup vs CPU software 609× speedup vs FATE/Paillier 229× faster HE addition AUC ≈ 0.79–0.88 (competitive) 200 MHz clock, DDR4 4-ch	First FPGA accelerator for industrial secure tree boosting. Replaces 3072-bit Paillier with EC-ElGamal for 128-bit security at lower cost. Deployed in Ant Group's AntChain Station for financial risk control. GRE engine handles bandwidth bottleneck effectively for high-dimensional data.
19	June 2019 IEEE Access	Lattice-based FHE (LWE) MP12-Trapdoor Gadget matrix (GSW) Masking scheme (MIFHE) NAND-circuit evaluation	No empirical dataset Theoretical/formal construction; security proved via DLWE hardness assumption	Shorter params vs ABB-IBE (~15× smaller mpk) Smaller noise growth rate vs CM-MIFHE Smaller modulus q vs prior MIFHE Formal parameter comparison (Table 1 & 2)	Proposes improved IBFHE and first efficient leveled MIFHE using MP12-trapdoor. Allows homomorphic operations across ciphertexts from different user identities. Scheme is leveled (bounded depth); bootstrapping without obfuscation remains open. Relevant for multi-party cloud computation.
20	2020 / IEEE 2021 IEEE ANTS 2020	ElGamal FHE Virus Contact Map (VCM) platform PrivacyContactTrace algorithm NIST randomness test suite	Synthetic spatiotemporal GPS 10 COVID-19 positive patients + 1 query person. 20×20 km region, 24 hr window. Based on Google Timeline data of authors.	Passes all 15 NIST statistical tests Proportion range: 0.964–0.988 5 contact scenarios detected correctly Secondary contacts (surface exposure) handled	Novel FHE-based contact tracing that executes spatial-temporal queries entirely on encrypted data – no decryption needed. Addresses both direct (simultaneous presence) and secondary (surface transmission) contacts. Integrated with VCM surveillance platform. Cryptographic randomness confirmed. HIPA

LITERATURE SUMMARY

- **Secure Cloud & Banking Integration:** Paillier allows cloud servers to process sensitive banking data without ever seeing the raw information. This ensures that privacy is maintained even if the cloud provider's environment is not fully trusted.

- **Additive Homomorphism:** Paillier is a public-key cryptosystem primarily valued for its additive homomorphic property. This allows two ciphertexts to be mathematically combined such that the result, when decrypted, is the exact sum of the original plaintexts.

- **Non-Deterministic Security:** The algorithm is probabilistic (non-deterministic), meaning the same plaintext will yield different ciphertexts each time it is encrypted. This effectively prevents "dictionary attacks" where an adversary tries to guess data by comparing encrypted patterns.

- **Extended Mathematical Operations:** Beyond simple addition, the system can be adapted to perform encrypted subtraction and multiplication of a ciphertext by a plaintext constant, enabling complex computations in untrusted environments.

- **Key application areas** include healthcare data analysis, secure financial transactions, blockchain smart contracts, cloud computing, and privacy-preserving machine learning.

Problem Statement

Data at Rest

Banking servers store account balances in plaintext on cloud databases — vulnerable to breaches.

Computation Requires Decryption

Every deposit, withdrawal, or transfer forces the server to decrypt data, exposing sensitive figures.

Insider Attacks

Database admins and server operators can read plaintext financial data during computation.

Solution: Perform all banking computations directly on ENCRYPTED data — without ever decrypting balances.

PROJECT OBJECTIVES

1

Implement Paillier

From scratch, the Paillier homomorphic encryption scheme.

2

Design Banking System

Operate on encrypted data for core banking functions.

3

Demonstrate Privacy

Practical privacy-preserving computations in banking scenarios.

4

User-Friendly Interface

Provide an intuitive interface for banking operations.

5

Evaluate System

Assess performance and security of the implemented system.

METHODOLOGY: PAILLIER HOMOMORPHIC ENCRYPTION

1. Key Generation:

   $n = p \times q$

   $\lambda = \mathrm{lcm}(p-1, q-1)$

   $\mu = (L(g^{\lambda} \bmod n^2))^{-1} \bmod n$

2. Encryption:

   $c = g^m \cdot r^n \bmod n^2$

   (Balances stored as ciphertext)

3. Homomorphic Operations:

Deposit: $E(new) = E(balance) \times E(amount) \bmod n^2$.

Withdraw: $E(new) = E(balance) \times E(n\text{-}amount) \bmod n^2$.

4. Transfer:

   Sender: $E(S') = E(S) \times E(n\text{-}amount) \bmod n^2$.

   Receiver: $E(R') = E(R) \times E(amount) \bmod n^2$.

5. Decryption (Authorized Only):

   $m = L(c^{\lambda} \bmod n^2) \times \mu \bmod n$

where,

p = First large prime number

q = Second large prime number

n= Public modulus used for encryption

λ = Private key parameter

μ = Modular inverse used in decryption

g = Generator (usually n+1)

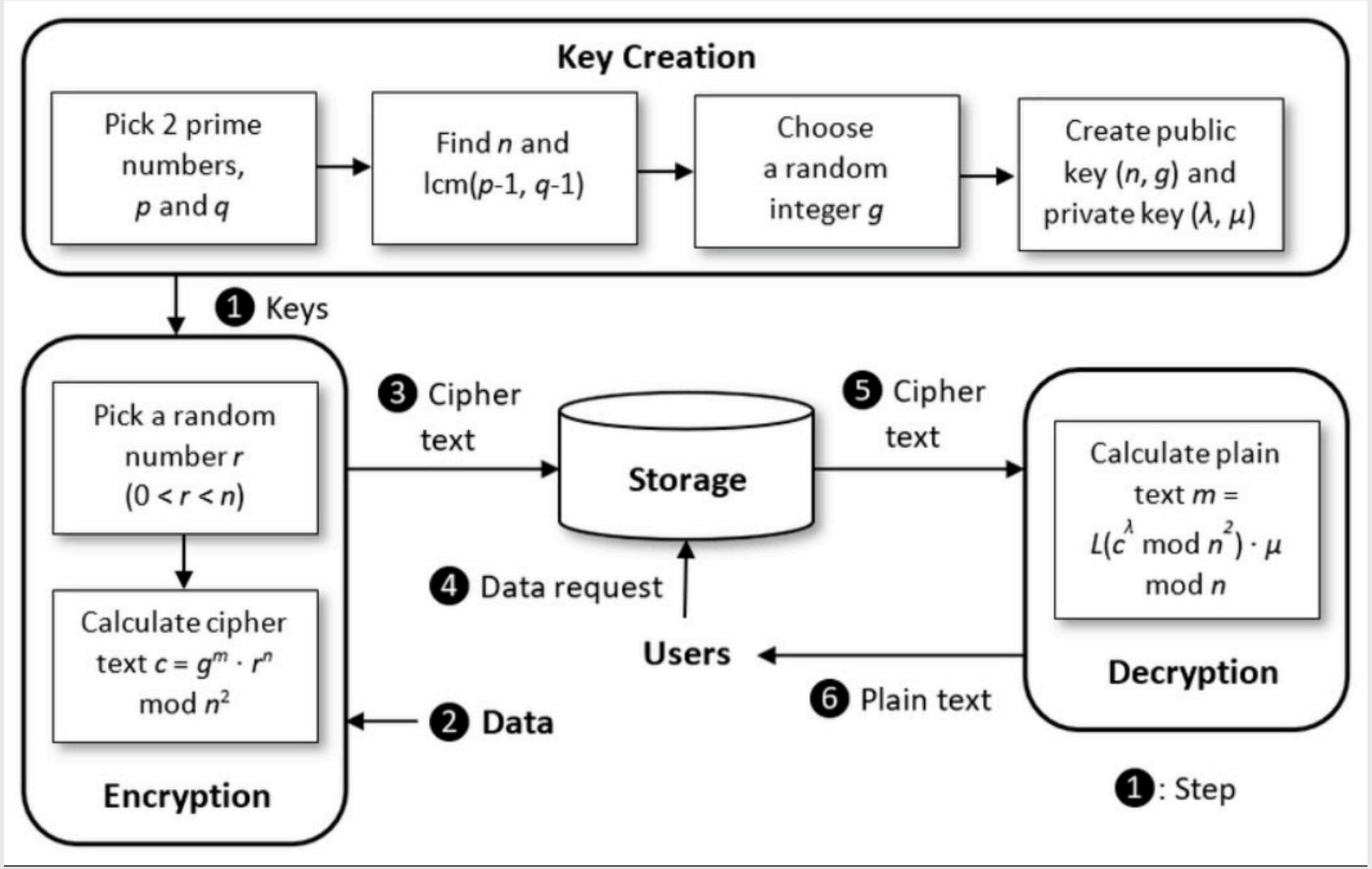
L(x)=(x-1)/n

c = Ciphertext

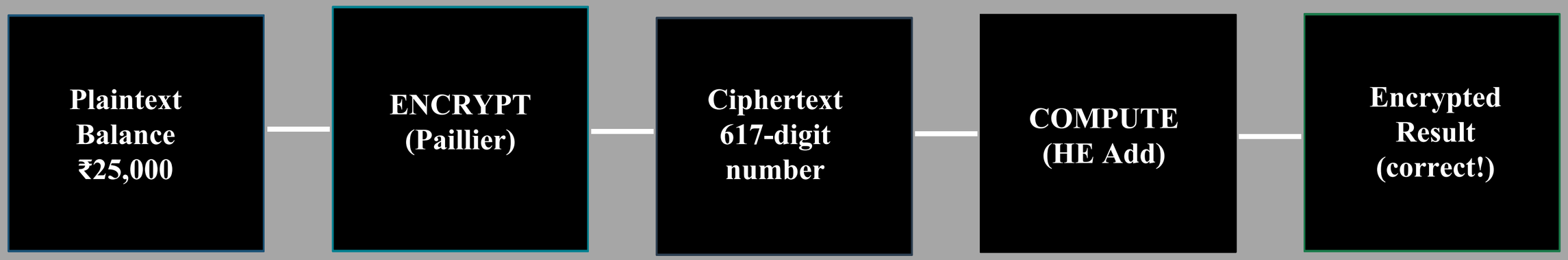
m = Plaintext value (account balance or amount)

r= Random integer (adds probabilistic security)

SYSTEM OVERVIEW



Homomorphic Encryption — Core Concept

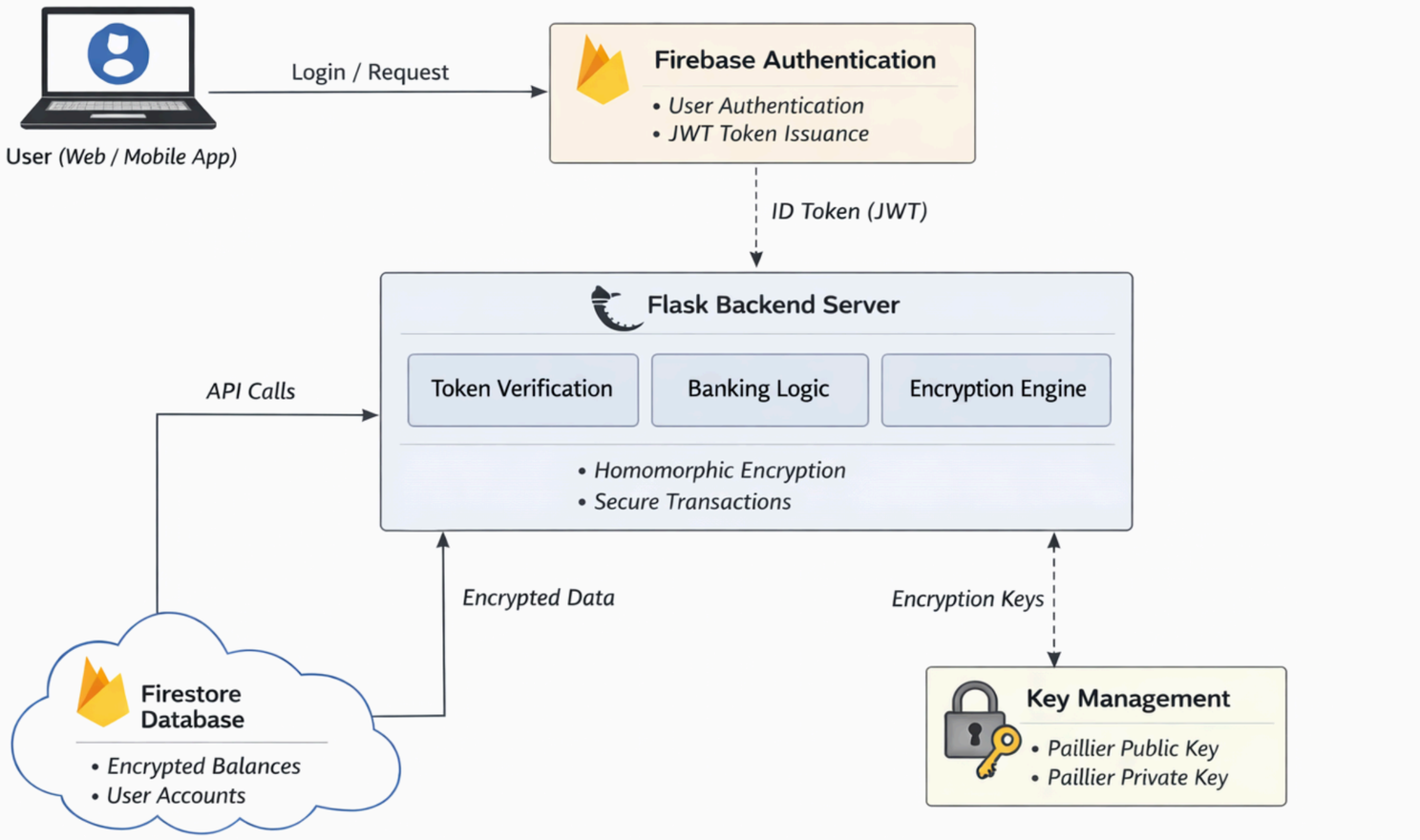


The Key Property:

$$Enc(a) \times Enc(b) \mod n^2 = Enc(a + b)$$

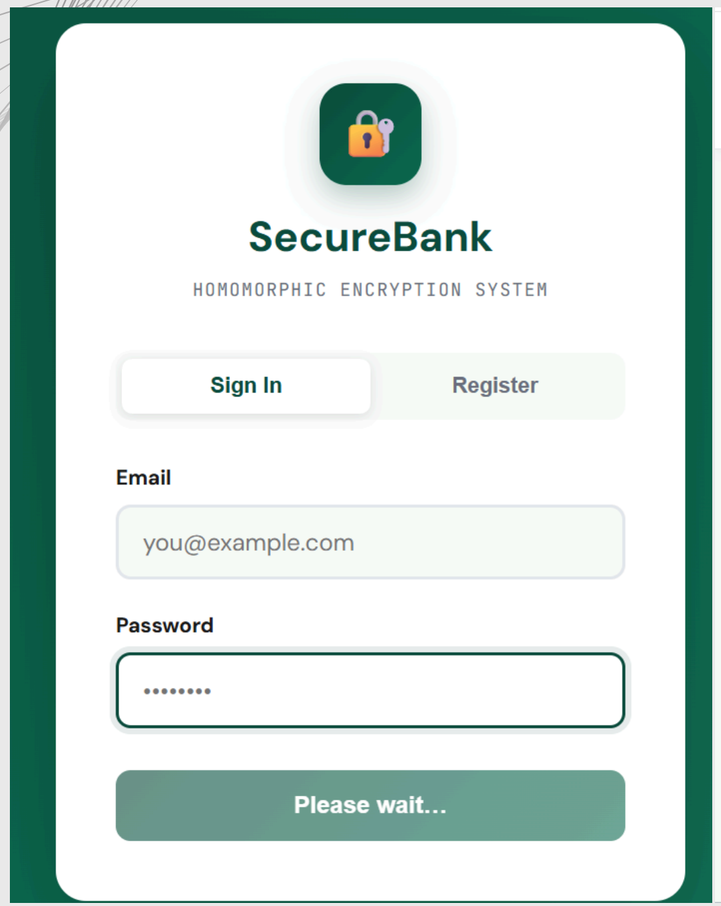
A server can add a deposit amount to an encrypted balance WITHOUT ever knowing the current balance.

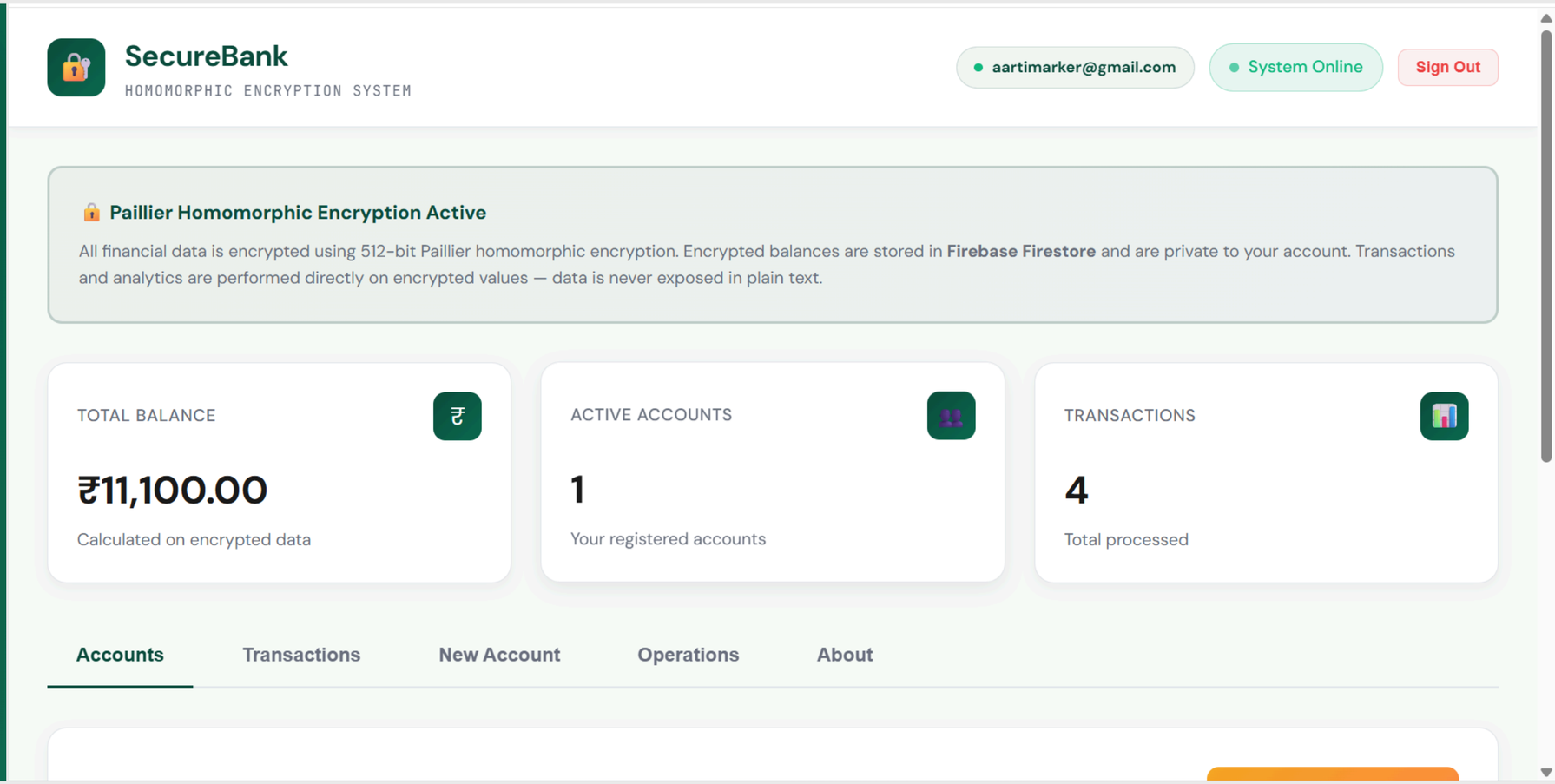
SYSTEM OVERVIEW



Secure Banking Data Privacy Preservation System with Homomorphic Encryption

USER INTERFACE





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CONCLUSION

- Privacy & Security: The system ensures data privacy by performing all banking operations (deposits, transfers, and analytics) directly on Paillier homomorphic ciphertexts, preventing Firebase administrators or external attackers from accessing plaintext balances.
- Practical Performance: Benchmarking confirms that Paillier HE is viable for real-world banking workloads, with end-to-end transaction latencies ranging from 29.4 ms to 73.5 ms, well within acceptable limits for standard financial APIs.
- Uncompromised Accuracy: Unlike other schemes, the Paillier implementation achieved a 100% pass rate with zero decryption error, proving that its exact integer arithmetic is superior to BFV for sensitive financial calculations.
- Storage & Efficiency: The system is highly resource-efficient, utilizing ciphertexts that are 60% smaller than BFV alternatives (0.60 KB vs 1.38 KB), which significantly reduces long-term storage costs in cloud databases like Firestore.
- Scalable Architecture: The project delivers a production-grade, full-stack solution using a Python-based API and Firebase/Firestore, providing a modular framework that can be easily integrated into existing horizontally scalable backends.

THANK YOU!